ML Architecture

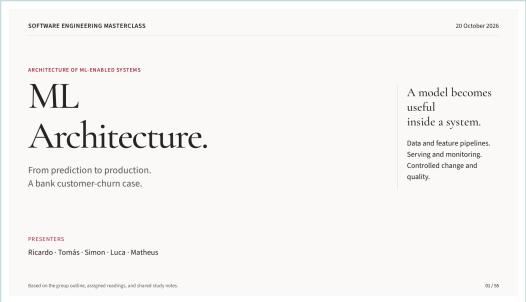
Speaker Notes

A bank customer-churn case — 20 October 2026

Each page shows one slide exactly as presenters will see it, the key points to cover, delivery cues, and a worked spoken-example paragraph to practice from. The example is a model to rehearse and adapt, not a transcript to read verbatim. Animated builds are marked; press the arrow key the stated number of times before moving to the next slide.

01 / 55 ML Architecture

Block 1 / Foundations • All presenters



CUES

- New block: **Block 1 / Foundations**.
- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- From a trained model to a maintainable, observable, governed system.
- **Software engineering** for systems that depend on learned behavior.
- **One shared case**: bank customer-churn prediction.
- **Two blocks**: model-to-system foundations, then architecture and quality.
- **Five presenters**: Ricardo, Tomás, Simon, Luca, and Matheus.
- Bank churn supplies the shared case. Examples, records, schedules, and incidents are illustrative.
- *A useful model needs an engineered system around it.*

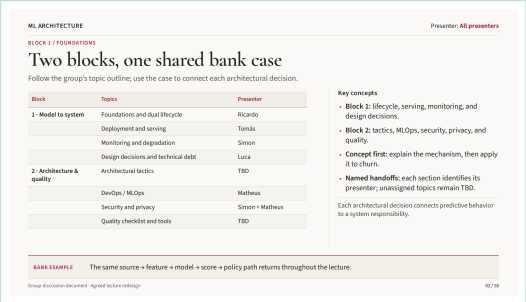
SPOKEN EXAMPLE — practice this out loud

From a trained model to a maintainable, observable, governed system. Software engineering means for systems that depend on learned behavior. One shared case means bank customer-churn prediction. Two blocks means model-to-system foundations, then architecture and quality. Five presenters means ricardo, Tomás, Simon, Luca, and Matheus. Bank churn supplies the shared case. Examples, records, schedules, and incidents are illustrative. A useful model needs an engineered system around it.

Reading anchor: Bass, Ch. 14 · Kalinowski, Ch. 13–14

02 / 55 Two blocks, one shared bank case

Block 1 / Foundations • All presenters



CUES

- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Follow the group's topic outline; use the case to connect each architectural decision.
- **Block 1**: lifecycle, serving, monitoring, and design decisions.
- **Block 2**: tactics, MLOps, security, privacy, and quality.
- **Concept first**: explain the mechanism, then apply it to churn.
- **Named handoffs**: each section identifies its presenter; unassigned topics remain TBD.
- *At our bank,* the same source → feature → model → score → policy path returns throughout the lecture.
- *Each architectural decision connects predictive behavior to a system responsibility.*

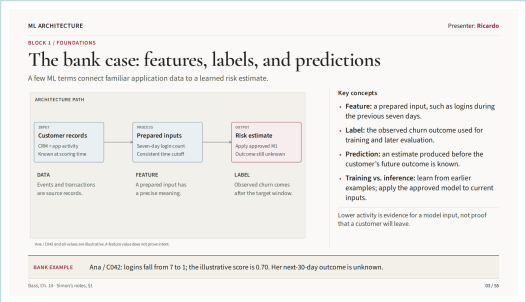
SPOKEN EXAMPLE — practice this out loud

Follow the group's topic outline; use the case to connect each architectural decision. Block 1 means lifecycle, serving, monitoring, and design decisions. Block 2 means tactics, MLOps, security, privacy, and quality. Concept first means explain the mechanism, then apply it to churn. Named handoffs means each section identifies its presenter; unassigned topics remain TBD. At our bank, the same source → feature → model → score → policy path returns throughout the lecture. Each architectural decision connects predictive behavior to a system responsibility.

Reading anchor: Group discussion document · Agreed lecture redesign

03 / 55 The bank case: features, labels, and predictions

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A few ML terms connect familiar application data to a learned risk estimate.
- **Feature**: a prepared input, such as logins during the previous seven days.
- **Label**: the observed churn outcome used for training and later evaluation.
- **Prediction**: an estimate produced before the customer's future outcome is known.
- **Training vs. inference**: learn from earlier examples; apply the approved model to current inputs.
- *At our bank,* ana / C042: logins fall from 7 to 1; the illustrative score is 0.70. Her next-30-day outcome is unknown.
- *Lower activity is evidence for a model input, not proof that a customer will leave.*

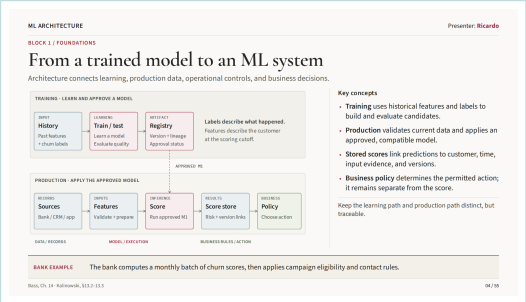
SPOKEN EXAMPLE — practice this out loud

A few ML terms connect familiar application data to a learned risk estimate. Feature means a prepared input, such as logins during the previous seven days. Label means the observed churn outcome used for training and later evaluation. Prediction means an estimate produced before the customer's future outcome is known. Training vs. inference means learn from earlier examples; apply the approved model to current inputs. At our bank, ana / C042: logins fall from 7 to 1; the illustrative score is 0.70. Her next-30-day outcome is unknown. Lower activity is evidence for a model input, not proof that a customer will leave.

Reading anchor: Bass, Ch. 14 · Simon's notes, §1

04 / 55 From a trained model to an ML system

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Architecture connects learning, production data, operational controls, and business decisions.
- **Training** uses historical features and labels to build and evaluate candidates.
- **Production** validates current data and applies an approved, compatible model.
- **Stored scores** link predictions to customer, time, input evidence, and versions.
- **Business policy** determines the permitted action; it remains separate from the score.
- *At our bank,* the bank computes a monthly batch of churn scores, then applies campaign eligibility and contact rules.
- *Keep the learning path and production path distinct, but traceable.*

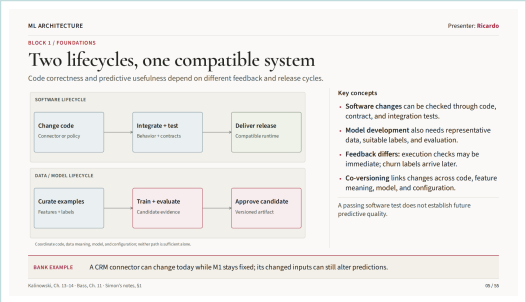
SPOKEN EXAMPLE — practice this out loud

Architecture connects learning, production data, operational controls, and business decisions. Training means uses historical features and labels to build and evaluate candidates. Production means validates current data and applies an approved, compatible model. Stored scores means link predictions to customer, time, input evidence, and versions. Business policy means determines the permitted action; it remains separate from the score. At our bank, the bank computes a monthly batch of churn scores, then applies campaign eligibility and contact rules. Keep the learning path and production path distinct, but traceable.

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.2–13.3

05 / 55 Two lifecycles, one compatible system

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Code correctness and predictive usefulness depend on different feedback and release cycles.
- **Software changes** can be checked through code, contract, and integration tests.
- **Model development** also needs representative data, suitable labels, and evaluation.
- **Feedback differs**: execution checks may be immediate; churn labels arrive later.
- **Co-versioning** links changes across code, feature meaning, model, and configuration.
- *At our bank,* a CRM connector can change today while M1 stays fixed; its changed inputs can still alter predictions.
- *A passing software test does not establish future predictive quality.*

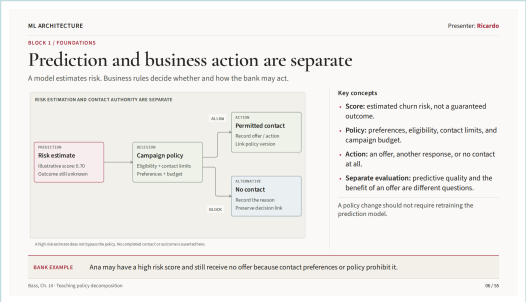
SPOKEN EXAMPLE — practice this out loud

Code correctness and predictive usefulness depend on different feedback and release cycles. Software changes means can be checked through code, contract, and integration tests. Model development means also needs representative data, suitable labels, and evaluation. Feedback differs means execution checks may be immediate; churn labels arrive later. Co-versioning means links changes across code, feature meaning, model, and configuration. At our bank, a CRM connector can change today while M1 stays fixed; its changed inputs can still alter predictions. A passing software test does not establish future predictive quality.

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 · Simon’s notes, §1

06 / 55 Prediction and business action are separate

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A model estimates risk. Business rules decide whether and how the bank may act.
- **Score**: estimated churn risk, not a guaranteed outcome.
- **Policy**: preferences, eligibility, contact limits, and campaign budget.
- **Action**: an offer, another response, or no contact at all.
- **Separate evaluation**: predictive quality and the benefit of an offer are different questions.
- *At our bank,* ana may have a high risk score and still receive no offer because contact preferences or policy prohibit it.
- *A policy change should not require retraining the prediction model.*

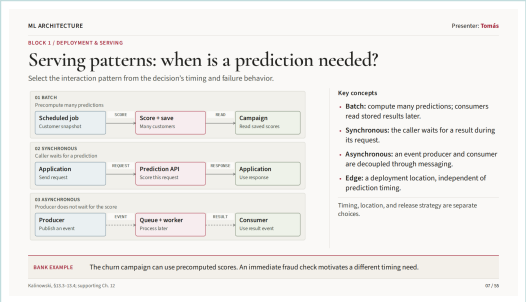
SPOKEN EXAMPLE — practice this out loud

A model estimates risk. Business rules decide whether and how the bank may act. Score means estimated churn risk, not a guaranteed outcome. Policy means preferences, eligibility, contact limits, and campaign budget. Action means an offer, another response, or no contact at all. Separate evaluation means predictive quality and the benefit of an offer are different questions. At our bank, ana may have a high risk score and still receive no offer because contact preferences or policy prohibit it. A policy change should not require retraining the prediction model.

Reading anchor: Bass, Ch. 14 · Teaching policy decomposition

07 / 55 Serving patterns: when is a prediction needed?

Block 1 / Deployment & serving • Tomás



CUES

- New block: **Block 1 / Deployment & serving**.
- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Select the interaction pattern from the decision's timing and failure behavior.
- **Batch**: compute many predictions; consumers read stored results later.
- **Synchronous**: the caller waits for a result during its request.
- **Asynchronous**: an event producer and consumer are decoupled through messaging.
- **Edge**: a deployment location, independent of prediction timing.
- *At our bank,* the churn campaign can use precomputed scores. An immediate fraud check motivates a different timing need.
- *Timing, location, and release strategy are separate choices.*

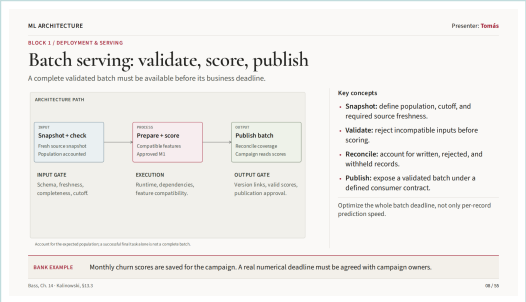
SPOKEN EXAMPLE — practice this out loud

Select the interaction pattern from the decision's timing and failure behavior. Batch means compute many predictions; consumers read stored results later. Synchronous means the caller waits for a result during its request. Asynchronous means an event producer and consumer are decoupled through messaging. Edge means a deployment location, independent of prediction timing. At our bank, the churn campaign can use precomputed scores. An immediate fraud check motivates a different timing need. Timing, location, and release strategy are separate choices.

Reading anchor: Kalinowski, §13.3–13.4; supporting Ch. 12

08 / 55 Batch serving: validate, score, publish

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A complete validated batch must be available before its business deadline.
- **Snapshot**: define population, cutoff, and required source freshness.
- **Validate**: reject incompatible inputs before scoring.
- **Reconcile**: account for written, rejected, and withheld records.
- **Publish**: expose a validated batch under a defined consumer contract.
- *At our bank,* monthly churn scores are saved for the campaign. A real numerical deadline must be agreed with campaign owners.
- *Optimize the whole batch deadline, not only per-record prediction speed.*

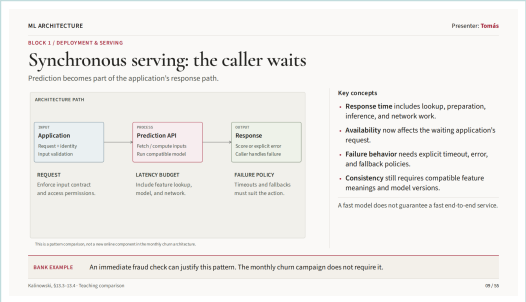
SPOKEN EXAMPLE — practice this out loud

A complete validated batch must be available before its business deadline. Snapshot means define population, cutoff, and required source freshness. Validate means reject incompatible inputs before scoring. Reconcile means account for written, rejected, and withheld records. Publish means expose a validated batch under a defined consumer contract. At our bank, monthly churn scores are saved for the campaign. A real numerical deadline must be agreed with campaign owners. Optimize the whole batch deadline, not only per-record prediction speed.

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3

09 / 55 Synchronous serving: the caller waits

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Prediction becomes part of the application’s response path.
- **Response time** includes lookup, preparation, inference, and network work.
- **Availability** now affects the waiting application’s request.
- **Failure behavior** needs explicit timeout, error, and fallback policies.
- **Consistency** still requires compatible feature meanings and model versions.
- *At our bank,* an immediate fraud check can justify this pattern. The monthly churn campaign does not require it.
- *A fast model does not guarantee a fast end-to-end service.*

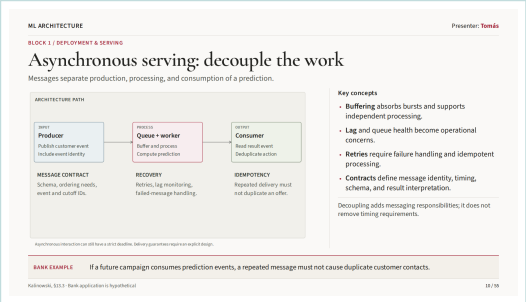
SPOKEN EXAMPLE — practice this out loud

Prediction becomes part of the application’s response path. Response time means includes lookup, preparation, inference, and network work. Availability means now affects the waiting application’s request. Failure behavior means needs explicit timeout, error, and fallback policies. Consistency means still requires compatible feature meanings and model versions. At our bank, an immediate fraud check can justify this pattern. The monthly churn campaign does not require it. A fast model does not guarantee a fast end-to-end service.

Reading anchor: Kalinowski, §13.3–13.4 · Teaching comparison

10 / 55 Asynchronous serving: decouple the work

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Messages separate production, processing, and consumption of a prediction.
- **Buffering** absorbs bursts and supports independent processing.
- **Lag** and queue health become operational concerns.
- **Retries** require failure handling and idempotent processing.
- **Contracts** define message identity, timing, schema, and result interpretation.
- *At our bank,* if a future campaign consumes prediction events, a repeated message must not cause duplicate customer contacts.
- *Decoupling adds messaging responsibilities; it does not remove timing requirements.*

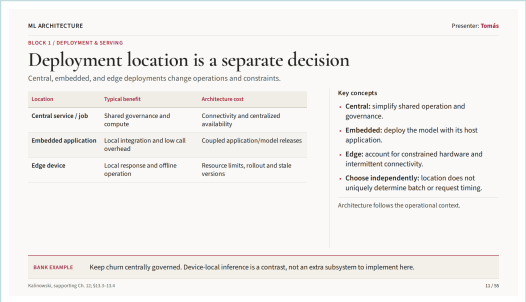
SPOKEN EXAMPLE — practice this out loud

Messages separate production, processing, and consumption of a prediction. Buffering means absorbs bursts and supports independent processing. Lag means and queue health become operational concerns. Retries means require failure handling and idempotent processing. Contracts means define message identity, timing, schema, and result interpretation. At our bank, if a future campaign consumes prediction events, a repeated message must not cause duplicate customer contacts. Decoupling adds messaging responsibilities; it does not remove timing requirements.

Reading anchor: Kalinowski, §13.3 · Bank application is hypothetical

11 / 55 Deployment location is a separate decision

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Central, embedded, and edge deployments change operations and constraints.
- **Central**: simplify shared operation and governance.
- **Embedded**: deploy the model with its host application.
- **Edge**: account for constrained hardware and intermittent connectivity.
- **Choose independently**: location does not uniquely determine batch or request timing.
- *At our bank,* keep churn centrally governed. Device-local inference is a contrast, not an extra subsystem to implement here.
- *Architecture follows the operational context.*

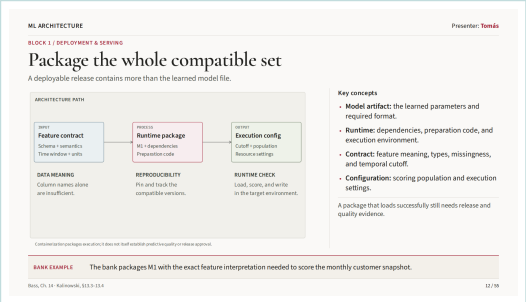
SPOKEN EXAMPLE — practice this out loud

Central, embedded, and edge deployments change operations and constraints. Central means simplify shared operation and governance. Embedded means deploy the model with its host application. Edge means account for constrained hardware and intermittent connectivity. Choose independently means location does not uniquely determine batch or request timing. At our bank, keep churn centrally governed. Device-local inference is a contrast, not an extra subsystem to implement here. Architecture follows the operational context.

Reading anchor: Kalinowski, supporting Ch. 12; §13.3–13.4

12 / 55 Package the whole compatible set

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A deployable release contains more than the learned model file.
- **Model artifact**: the learned parameters and required format.
- **Runtime**: dependencies, preparation code, and execution environment.
- **Contract**: feature meaning, types, missingness, and temporal cutoff.
- **Configuration**: scoring population and execution settings.
- *At our bank,* the bank packages M1 with the exact feature interpretation needed to score the monthly customer snapshot.
- *A package that loads successfully still needs release and quality evidence.*

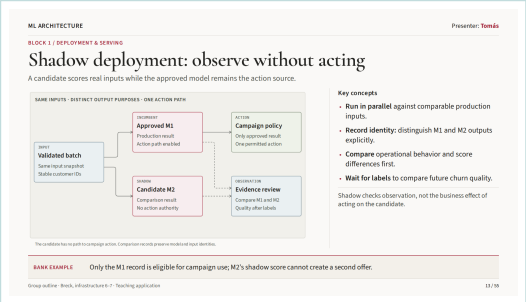
SPOKEN EXAMPLE — practice this out loud

A deployable release contains more than the learned model file. Model artifact means the learned parameters and required format. Runtime means dependencies, preparation code, and execution environment. Contract means feature meaning, types, missingness, and temporal cutoff. Configuration means scoring population and execution settings. At our bank, the bank packages M1 with the exact feature interpretation needed to score the monthly customer snapshot. A package that loads successfully still needs release and quality evidence.

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3–13.4

13 / 55 Shadow deployment: observe without acting

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A candidate scores real inputs while the approved model remains the action source.
- **Run in parallel** against comparable production inputs.
- **Record identity**: distinguish M1 and M2 outputs explicitly.
- **Compare** operational behavior and score differences first.
- **Wait for labels** to compare future churn quality.
- *At our bank,* only the M1 record is eligible for campaign use; M2's shadow score cannot create a second offer.
- *Shadow checks observation, not the business effect of acting on the candidate.*

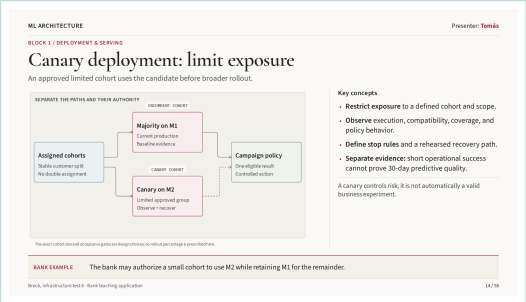
SPOKEN EXAMPLE — practice this out loud

A candidate scores real inputs while the approved model remains the action source. Run in parallel means against comparable production inputs. Record identity means distinguish M1 and M2 outputs explicitly. Compare means operational behavior and score differences first. Wait for labels means to compare future churn quality. At our bank, only the M1 record is eligible for campaign use; M2's shadow score cannot create a second offer. Shadow checks observation, not the business effect of acting on the candidate.

Reading anchor: Group outline · Breck, infrastructure 6–7 · Teaching application

14 / 55 Canary deployment: limit exposure

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- An approved limited cohort uses the candidate before broader rollout.
- **Restrict exposure** to a defined cohort and scope.
- **Observe** execution, compatibility, coverage, and policy behavior.
- **Define stop rules** and a rehearsed recovery path.
- **Separate evidence**: short operational success cannot prove 30-day predictive quality.
- *At our bank,* the bank may authorize a small cohort to use M2 while retaining M1 for the remainder.
- *A canary controls risk; it is not automatically a valid business experiment.*

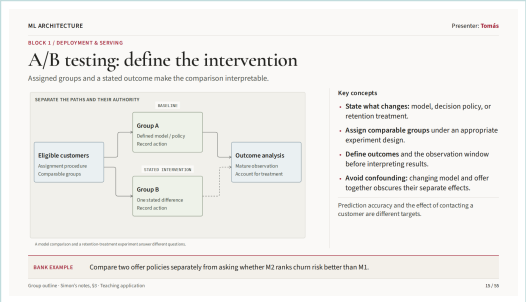
SPOKEN EXAMPLE — practice this out loud

An approved limited cohort uses the candidate before broader rollout. Restrict exposure means to a defined cohort and scope. Observe means execution, compatibility, coverage, and policy behavior. Define stop rules means and a rehearsed recovery path. Separate evidence means short operational success cannot prove 30-day predictive quality. At our bank, the bank may authorize a small cohort to use M2 while retaining M1 for the remainder. A canary controls risk; it is not automatically a valid business experiment.

Reading anchor: Breck, infrastructure test 6 · Bank teaching application

15 / 55 A/B testing: define the intervention

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Assigned groups and a stated outcome make the comparison interpretable.
- **State what changes**: model, decision policy, or retention treatment.
- **Assign comparable groups** under an appropriate experiment design.
- **Define outcomes** and the observation window before interpreting results.
- **Avoid confounding**: changing model and offer together obscures their separate effects.
- *At our bank,* compare two offer policies separately from asking whether M2 ranks churn risk better than M1.
- *Prediction accuracy and the effect of contacting a customer are different targets.*

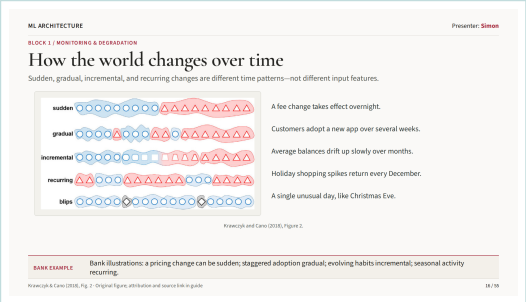
SPOKEN EXAMPLE — practice this out loud

Assigned groups and a stated outcome make the comparison interpretable. State what changes means model, decision policy, or retention treatment. Assign comparable groups means under an appropriate experiment design. Define outcomes means and the observation window before interpreting results. Avoid confounding means changing model and offer together obscures their separate effects. At our bank, compare two offer policies separately from asking whether M2 ranks churn risk better than M1. Prediction accuracy and the effect of contacting a customer are different targets.

Reading anchor: Group outline · Simon’s notes, §3 · Teaching application

16 / 55 How the world changes over time

Block 1 / Monitoring & degradation • Simon



CUES

- New block: **Block 1 / Monitoring & degradation**.
- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.
- Adaptation note on file for this slide; do not read aloud unless asked.

TO SAY

- Sudden, gradual, incremental, and recurring changes are different time patterns—not different input features.
- **Sudden**: the old regime switches abruptly.
- **Gradual**: old and new regimes coexist during a transition.
- **Incremental / recurring**: intermediate change differs from a returning regime.
- **Blips**: isolated noise is not necessarily a lasting regime change.
- Bank illustrations: a pricing change can be sudden; staggered adoption gradual; evolving habits incremental; seasonal activity recurring.
- *First ask how change unfolds over time. Then ask what changed: input data, outcome relationships, or data validity.*

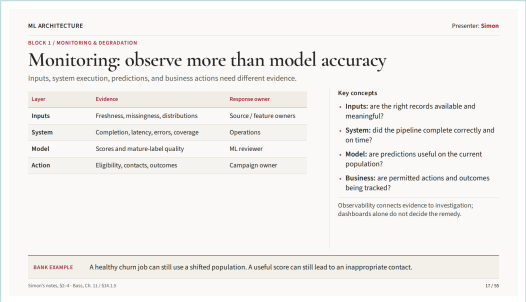
SPOKEN EXAMPLE — practice this out loud

Sudden, gradual, incremental, and recurring changes are different time patterns—not different input features. Sudden means the old regime switches abruptly. Gradual means old and new regimes coexist during a transition. Incremental / recurring means intermediate change differs from a returning regime. Blips means isolated noise is not necessarily a lasting regime change. Bank illustrations: a pricing change can be sudden; staggered adoption gradual; evolving habits incremental; seasonal activity recurring. First ask how change unfolds over time. Then ask what changed: input data, outcome relationships, or data validity.

Reading anchor: Krawczyk & Cano (2018), Fig. 2 · Original figure; attribution and source link in guide

17 / 55 Monitoring: observe more than model accuracy

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Inputs, system execution, predictions, and business actions need different evidence.
- **Inputs**: are the right records available and meaningful?
- **System**: did the pipeline complete correctly and on time?
- **Model**: are predictions useful on the current population?
- **Business**: are permitted actions and outcomes being tracked?
- *At our bank,* a healthy churn job can still use a shifted population. A useful score can still lead to an inappropriate contact.
- *Observability connects evidence to investigation; dashboards alone do not decide the remedy.*

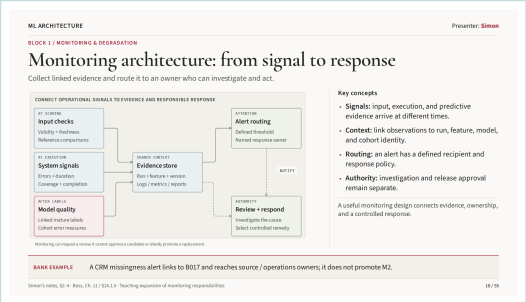
SPOKEN EXAMPLE — practice this out loud

Inputs, system execution, predictions, and business actions need different evidence. Inputs means are the right records available and meaningful? System means did the pipeline complete correctly and on time? Model means are predictions useful on the current population? Business means are permitted actions and outcomes being tracked? At our bank, a healthy churn job can still use a shifted population. A useful score can still lead to an inappropriate contact. Observability connects evidence to investigation; dashboards alone do not decide the remedy.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

18 / 55 Monitoring architecture: from signal to response

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Collect linked evidence and route it to an owner who can investigate and act.
- **Signals**: input, execution, and predictive evidence arrive at different times.
- **Context**: link observations to run, feature, model, and cohort identity.
- **Routing**: an alert has a defined recipient and response policy.
- **Authority**: investigation and release approval remain separate.
- *At our bank,* a CRM missingness alert links to B017 and reaches source / operations owners; it does not promote M2.
- *A useful monitoring design connects evidence, ownership, and a controlled response.*

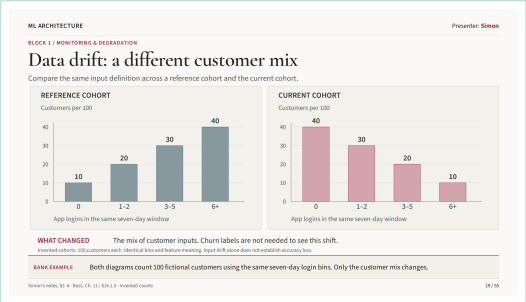
SPOKEN EXAMPLE — practice this out loud

Collect linked evidence and route it to an owner who can investigate and act. Signals means input, execution, and predictive evidence arrive at different times. Context means link observations to run, feature, model, and cohort identity. Routing means an alert has a defined recipient and response policy. Authority means investigation and release approval remain separate. At our bank, a CRM missingness alert links to B017 and reaches source / operations owners; it does not promote M2. A useful monitoring design connects evidence, ownership, and a controlled response.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching expansion of monitoring responsibilities

19 / 55 Data drift: a different customer mix

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Compare the same input definition across a reference cohort and the current cohort.
- **What changes**: the distribution of valid model inputs.
- **Bank example**: a campaign attracts more low-activity customers.
- **Available now**: compare inputs without waiting for churn labels.
- **Limit**: a changed distribution alone does not prove lost accuracy.
- *At our bank,* both diagrams count 100 fictional customers using the same seven-day login bins. Only the customer mix changes.
- *Choose a meaningful reference, comparable cohorts, and a defined response.*

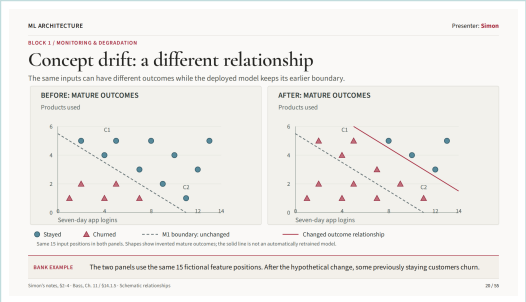
SPOKEN EXAMPLE — practice this out loud

Compare the same input definition across a reference cohort and the current cohort. What changes means the distribution of valid model inputs. Bank example means a campaign attracts more low-activity customers. Available now means compare inputs without waiting for churn labels. Limit means a changed distribution alone does not prove lost accuracy. At our bank, both diagrams count 100 fictional customers using the same seven-day login bins. Only the customer mix changes. Choose a meaningful reference, comparable cohorts, and a defined response.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented counts

20 / 55 Concept drift: a different relationship

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- The same inputs can have different outcomes while the deployed model keeps its earlier boundary.
- **What changes**: churn risk given the same valid input information.
- **Bank example**: new fees make active customers leave more often.
- **Evidence**: link predictions to mature, reliable churn outcomes.
- **Model status**: M1 stays fixed until a separately evaluated release changes it.
- *At our bank,* the two panels use the same 15 fictional feature positions. After the hypothetical change, some previously staying customers churn.
- *A missing field is not concept drift; a statistical alarm is not a retraining approval.*

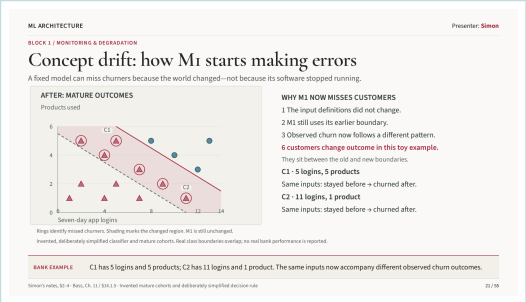
SPOKEN EXAMPLE — practice this out loud

The same inputs can have different outcomes while the deployed model keeps its earlier boundary. What changes means churn risk given the same valid input information. Bank example means new fees make active customers leave more often. Evidence means link predictions to mature, reliable churn outcomes. Model status means m1 stays fixed until a separately evaluated release changes it. At our bank, the two panels use the same 15 fictional feature positions. After the hypothetical change, some previously staying customers churn. A missing field is not concept drift; a statistical alarm is not a retraining approval.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Schematic relationships

21 / 55 Concept drift: how M1 starts making errors

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.
- Adaptation note on file for this slide; do not read aloud unless asked.

TO SAY

- A fixed model can miss churners because the world changed—not because its software stopped running.
- **Inputs stay fixed**: the two cohorts use identical feature positions.
- **M1 stays fixed**: its earlier decision boundary is unchanged.
- **Outcomes change**: a different relationship creates missed churners.
- **Evidence**: the example uses mature labels, not outcomes already known on scoring day.
- *At our bank,* c1 has 5 logins and 5 products; C2 has 11 logins and 1 product. The same inputs now accompany different observed churn outcomes.
- *Changed labels reveal the mismatch. Investigate causes, evaluate candidates, and use controlled release rather than automatic promotion.*

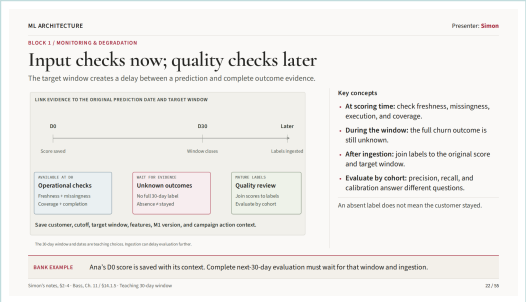
SPOKEN EXAMPLE — practice this out loud

A fixed model can miss churners because the world changed—not because its software stopped running. Inputs stay fixed means the two cohorts use identical feature positions. M1 stays fixed means its earlier decision boundary is unchanged. Outcomes change means a different relationship creates missed churners. Evidence means the example uses mature labels, not outcomes already known on scoring day. At our bank, c1 has 5 logins and 5 products; C2 has 11 logins and 1 product. The same inputs now accompany different observed churn outcomes. Changed labels reveal the mismatch. Investigate causes, evaluate candidates, and use controlled release rather than automatic promotion.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented mature cohorts and deliberately simplified decision rule

22 / 55 Input checks now; quality checks later

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- The target window creates a delay between a prediction and complete outcome evidence.
- **At scoring time**: check freshness, missingness, execution, and coverage.
- **During the window**: the full churn outcome is still unknown.
- **After ingestion**: join labels to the original score and target window.
- **Evaluate by cohort**: precision, recall, and calibration answer different questions.
- *At our bank,* ana's D0 score is saved with its context. Complete next-30-day evaluation must wait for that window and ingestion.
- *An absent label does not mean the customer stayed.*

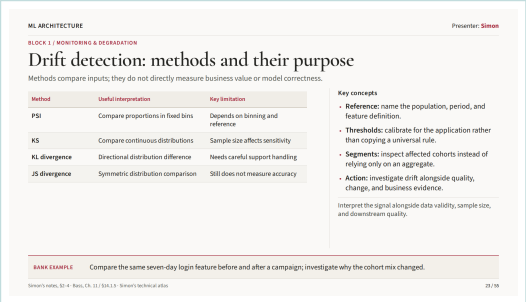
SPOKEN EXAMPLE — practice this out loud

The target window creates a delay between a prediction and complete outcome evidence. At scoring time means check freshness, missingness, execution, and coverage. During the window means the full churn outcome is still unknown. After ingestion means join labels to the original score and target window. Evaluate by cohort means precision, recall, and calibration answer different questions. At our bank, ana's D0 score is saved with its context. Complete next-30-day evaluation must wait for that window and ingestion. An absent label does not mean the customer stayed.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching 30-day window

23 / 55 Drift detection: methods and their purpose

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Methods compare inputs; they do not directly measure business value or model correctness.
- **Reference**: name the population, period, and feature definition.
- **Thresholds**: calibrate for the application rather than copying a universal rule.
- **Segments**: inspect affected cohorts instead of relying only on an aggregate.
- **Action**: investigate drift alongside quality, change, and business evidence.
- *At our bank,* compare the same seven-day login feature before and after a campaign; investigate why the cohort mix changed.
- *Interpret the signal alongside data validity, sample size, and downstream quality.*

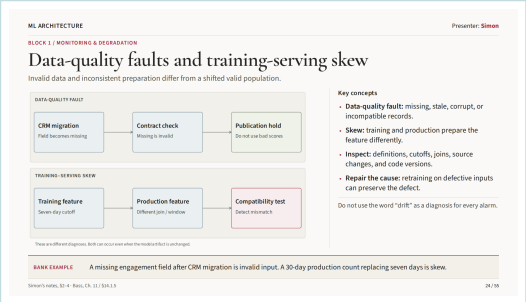
SPOKEN EXAMPLE — practice this out loud

Methods compare inputs; they do not directly measure business value or model correctness. Reference means name the population, period, and feature definition. Thresholds means calibrate for the application rather than copying a universal rule. Segments means inspect affected cohorts instead of relying only on an aggregate. Action means investigate drift alongside quality, change, and business evidence. At our bank, compare the same seven-day login feature before and after a campaign; investigate why the cohort mix changed. Interpret the signal alongside data validity, sample size, and downstream quality.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Simon’s technical atlas

24 / 55 Data-quality faults and training-serving skew

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Invalid data and inconsistent preparation differ from a shifted valid population.
- **Data-quality fault**: missing, stale, corrupt, or incompatible records.
- **Skew**: training and production prepare the feature differently.
- **Inspect**: definitions, cutoffs, joins, source changes, and code versions.
- **Repair the cause**: retraining on defective inputs can preserve the defect.
- *At our bank,* a missing engagement field after CRM migration is invalid input. A 30-day production count replacing seven days is skew.
- *Do not use the word “drift” as a diagnosis for every alarm.*

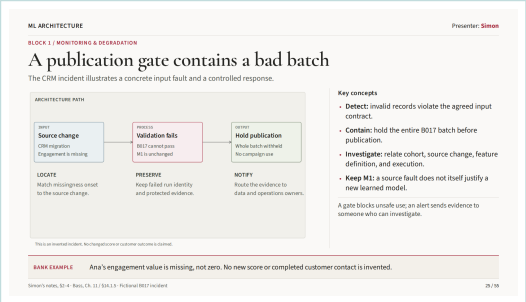
SPOKEN EXAMPLE — practice this out loud

Invalid data and inconsistent preparation differ from a shifted valid population. Data-quality fault means missing, stale, corrupt, or incompatible records. Skew means training and production prepare the feature differently. Inspect means definitions, cutoffs, joins, source changes, and code versions. Repair the cause means retraining on defective inputs can preserve the defect. At our bank, a missing engagement field after CRM migration is invalid input. A 30-day production count replacing seven days is skew. Do not use the word “drift” as a diagnosis for every alarm.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

25 / 55 A publication gate contains a bad batch

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- The CRM incident illustrates a concrete input fault and a controlled response.
- **Detect**: invalid records violate the agreed input contract.
- **Contain**: hold the entire B017 batch before publication.
- **Investigate**: relate cohort, source change, feature definition, and execution.
- **Keep M1**: a source fault does not itself justify a new learned model.
- *At our bank,* ana's engagement value is missing, not zero. No new score or completed customer contact is invented.
- *A gate blocks unsafe use; an alert sends evidence to someone who can investigate.*

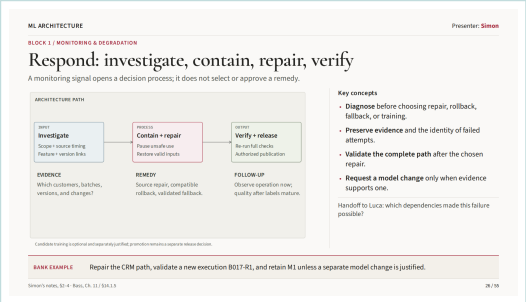
SPOKEN EXAMPLE — practice this out loud

The CRM incident illustrates a concrete input fault and a controlled response. Detect means invalid records violate the agreed input contract. Contain means hold the entire B017 batch before publication. Investigate means relate cohort, source change, feature definition, and execution. Keep M1 means a source fault does not itself justify a new learned model. At our bank, ana's engagement value is missing, not zero. No new score or completed customer contact is invented. A gate blocks unsafe use; an alert sends evidence to someone who can investigate.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Fictional B017 incident

26 / 55 Respond: investigate, contain, repair, verify

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A monitoring signal opens a decision process; it does not select or approve a remedy.
- **Diagnose** before choosing repair, rollback, fallback, or training.
- **Preserve evidence** and the identity of failed attempts.
- **Validate the complete path** after the chosen repair.
- **Request a model change** only when evidence supports one.
- *At our bank,* repair the CRM path, validate a new execution B017-R1, and retain M1 unless a separate model change is justified.
- *Handoff to Luca: which dependencies made this failure possible?*

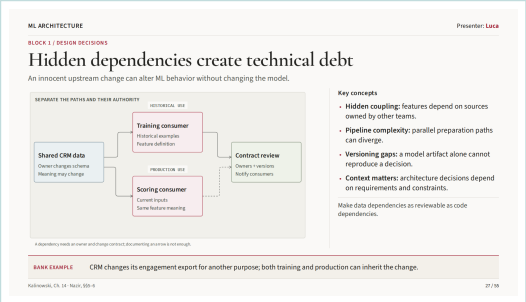
SPOKEN EXAMPLE — practice this out loud

A monitoring signal opens a decision process; it does not select or approve a remedy. Diagnose means before choosing repair, rollback, fallback, or training. Preserve evidence means and the identity of failed attempts. Validate the complete path means after the chosen repair. Request a model change means only when evidence supports one. At our bank, repair the CRM path, validate a new execution B017-R1, and retain M1 unless a separate model change is justified. Handoff to Luca: which dependencies made this failure possible?

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

27 / 55 Hidden dependencies create technical debt

Block 1 / Design decisions • Luca



CUES

- New block: **Block 1 / Design decisions**.
- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- An innocent upstream change can alter ML behavior without changing the model.
- **Hidden coupling**: features depend on sources owned by other teams.
- **Pipeline complexity**: parallel preparation paths can diverge.
- **Versioning gaps**: a model artifact alone cannot reproduce a decision.
- **Context matters**: architecture decisions depend on requirements and constraints.
- *At our bank,* cRM changes its engagement export for another purpose; both training and production can inherit the change.
- *Make data dependencies as reviewable as code dependencies.*

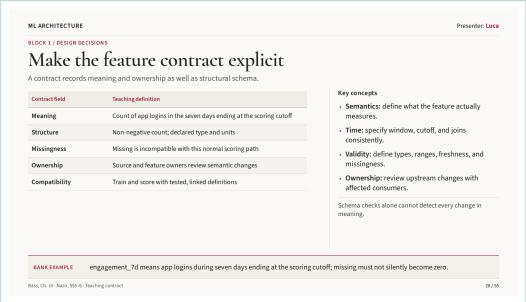
SPOKEN EXAMPLE — practice this out loud

An innocent upstream change can alter ML behavior without changing the model. Hidden coupling means features depend on sources owned by other teams. Pipeline complexity means parallel preparation paths can diverge. Versioning gaps means a model artifact alone cannot reproduce a decision. Context matters means architecture decisions depend on requirements and constraints. At our bank, cRM changes its engagement export for another purpose; both training and production can inherit the change. Make data dependencies as reviewable as code dependencies.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

28 / 55 Make the feature contract explicit

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A contract records meaning and ownership as well as structural schema.
- **Semantics**: define what the feature actually measures.
- **Time**: specify window, cutoff, and joins consistently.
- **Validity**: define types, ranges, freshness, and missingness.
- **Ownership**: review upstream changes with affected consumers.
- *At our bank,* engagement_7d means app logins during seven days ending at the scoring cutoff; missing must not silently become zero.
- *Schema checks alone cannot detect every change in meaning.*

SPOKEN EXAMPLE — practice this out loud

A contract records meaning and ownership as well as structural schema. Semantics means define what the feature actually measures. Time means specify window, cutoff, and joins consistently. Validity means define types, ranges, freshness, and missingness. Ownership means review upstream changes with affected consumers. At our bank, engagement_7d means app logins during seven days ending at the scoring cutoff; missing must not silently become zero. Schema checks alone cannot detect every change in meaning.

Reading anchor: Bass, Ch. 14 · Nazir, §§5–6 · Teaching contract

29 / 55 Separate responsibilities where change happens

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Clear interfaces let different concerns evolve without unnecessary relearning.
- **Single responsibility**: identify each module's reason to change.
- **Interfaces**: specify data and behavior at each boundary.
- **Dependency injection**: provide implementations through explicit contracts.
- **Deployment**: logical modules do not automatically require separate services.
- *At our bank,* a connector fix changes preparation; a contact-frequency rule changes campaign policy; a new M2 changes inference.
- *Separate concerns first; choose deployment units from the actual trade-offs.*

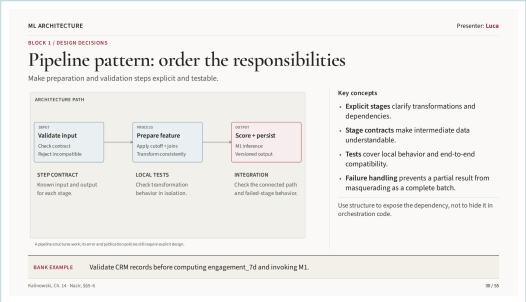
SPOKEN EXAMPLE — practice this out loud

Clear interfaces let different concerns evolve without unnecessary relearning. Single responsibility means identify each module's reason to change. Interfaces means specify data and behavior at each boundary. Dependency injection means provide implementations through explicit contracts. Deployment means logical modules do not automatically require separate services. At our bank, a connector fix changes preparation; a contact-frequency rule changes campaign policy; a new M2 changes inference. Separate concerns first; choose deployment units from the actual trade-offs.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

30 / 55 Pipeline pattern: order the responsibilities

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Make preparation and validation steps explicit and testable.
- **Explicit stages** clarify transformations and dependencies.
- **Stage contracts** make intermediate data understandable.
- **Tests** cover local behavior and end-to-end compatibility.
- **Failure handling** prevents a partial result from masquerading as a complete batch.
- *At our bank,* validate CRM records before computing engagement_7d and invoking M1.
- *Use structure to expose the dependency, not to hide it in orchestration code.*

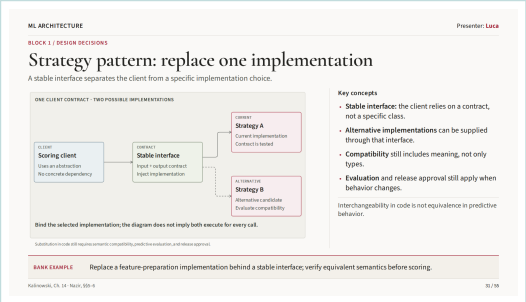
SPOKEN EXAMPLE — practice this out loud

Make preparation and validation steps explicit and testable. Explicit stages means clarify transformations and dependencies. Stage contracts means make intermediate data understandable. Tests means cover local behavior and end-to-end compatibility. Failure handling means prevents a partial result from masquerading as a complete batch. At our bank, validate CRM records before computing engagement_7d and invoking M1. Use structure to expose the dependency, not to hide it in orchestration code.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

31 / 55 Strategy pattern: replace one implementation

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A stable interface separates the client from a specific implementation choice.
- **Stable interface**: the client relies on a contract, not a specific class.
- **Alternative implementations** can be supplied through that interface.
- **Compatibility** still includes meaning, not only types.
- **Evaluation** and release approval still apply when behavior changes.
- *At our bank,* replace a feature-preparation implementation behind a stable interface; verify equivalent semantics before scoring.
- *Interchangeability in code is not equivalence in predictive behavior.*

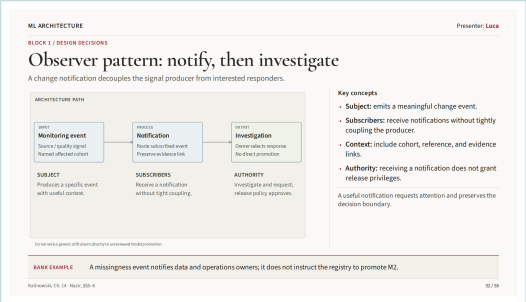
SPOKEN EXAMPLE — practice this out loud

A stable interface separates the client from a specific implementation choice. Stable interface means the client relies on a contract, not a specific class. Alternative implementations means can be supplied through that interface. Compatibility means still includes meaning, not only types. Evaluation means and release approval still apply when behavior changes. At our bank, replace a feature-preparation implementation behind a stable interface; verify equivalent semantics before scoring. Interchangeability in code is not equivalence in predictive behavior.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

32 / 55 Observer pattern: notify, then investigate

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A change notification decouples the signal producer from interested responders.
- **Subject**: emits a meaningful change event.
- **Subscribers**: receive notifications without tightly coupling the producer.
- **Context**: include cohort, reference, and evidence links.
- **Authority**: receiving a notification does not grant release privileges.
- *At our bank,* a missingness event notifies data and operations owners; it does not instruct the registry to promote M2.
- *A useful notification requests attention and preserves the decision boundary.*

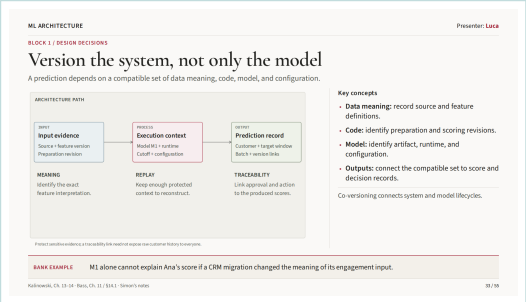
SPOKEN EXAMPLE — practice this out loud

A change notification decouples the signal producer from interested responders. Subject means emits a meaningful change event. Subscribers means receive notifications without tightly coupling the producer. Context means include cohort, reference, and evidence links. Authority means receiving a notification does not grant release privileges. At our bank, a missingness event notifies data and operations owners; it does not instruct the registry to promote M2. A useful notification requests attention and preserves the decision boundary.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

33 / 55 Version the system, not only the model

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A prediction depends on a compatible set of data meaning, code, model, and configuration.
- **Data meaning**: record source and feature definitions.
- **Code**: identify preparation and scoring revisions.
- **Model**: identify artifact, runtime, and configuration.
- **Outputs**: connect the compatible set to score and decision records.
- *At our bank,* m1 alone cannot explain Ana’s score if a CRM migration changed the meaning of its engagement input.
- *Co-versioning connects system and model lifecycles.*

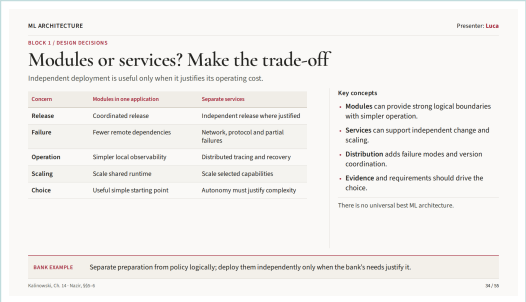
SPOKEN EXAMPLE — practice this out loud

A prediction depends on a compatible set of data meaning, code, model, and configuration. Data meaning means record source and feature definitions. Code means identify preparation and scoring revisions. Model means identify artifact, runtime, and configuration. Outputs means connect the compatible set to score and decision records. At our bank, m1 alone cannot explain Ana’s score if a CRM migration changed the meaning of its engagement input. Co-versioning connects system and model lifecycles.

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 / §14.1 · Simon’s notes

34 / 55 Modules or services? Make the trade-off

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Independent deployment is useful only when it justifies its operating cost.
- **Modules** can provide strong logical boundaries with simpler operation.
- **Services** can support independent change and scaling.
- **Distribution** adds failure modes and version coordination.
- **Evidence** and requirements should drive the choice.
- *At our bank,* separate preparation from policy logically; deploy them independently only when the bank's needs justify it.
- *There is no universal best ML architecture.*

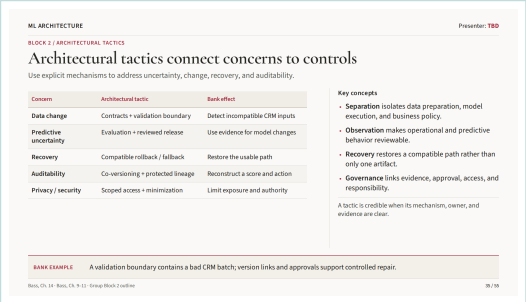
SPOKEN EXAMPLE — practice this out loud

Independent deployment is useful only when it justifies its operating cost. Modules means can provide strong logical boundaries with simpler operation. Services means can support independent change and scaling. Distribution means adds failure modes and version coordination. Evidence means and requirements should drive the choice. At our bank, separate preparation from policy logically; deploy them independently only when the bank's needs justify it. There is no universal best ML architecture.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

35 / 55 Architectural tactics connect concerns to controls

Block 2 / Architectural tactics • TBD



CUES

- New block: **Block 2 / Architectural tactics**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Use explicit mechanisms to address uncertainty, change, recovery, and auditability.
- **Separation** isolates data preparation, model execution, and business policy.
- **Observation** makes operational and predictive behavior reviewable.
- **Recovery** restores a compatible path rather than only one artifact.
- **Governance** links evidence, approval, access, and responsibility.
- *At our bank,* a validation boundary contains a bad CRM batch; version links and approvals support controlled repair.
- *A tactic is credible when its mechanism, owner, and evidence are clear.*

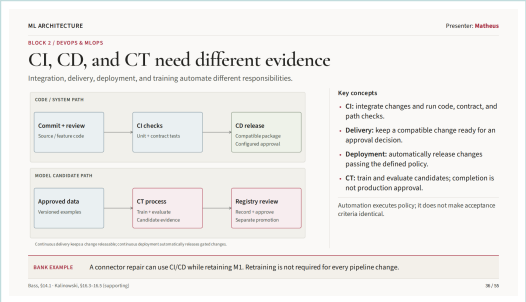
SPOKEN EXAMPLE — practice this out loud

Use explicit mechanisms to address uncertainty, change, recovery, and auditability. Separation means isolates data preparation, model execution, and business policy. Observation means makes operational and predictive behavior reviewable. Recovery means restores a compatible path rather than only one artifact. Governance means links evidence, approval, access, and responsibility. At our bank, a validation boundary contains a bad CRM batch; version links and approvals support controlled repair. A tactic is credible when its mechanism, owner, and evidence are clear.

Reading anchor: Bass, Ch. 14 · Bass, Ch. 9–11 · Group Block 2 outline

36 / 55 CI, CD, and CT need different evidence

Block 2 / DevOps & MLOps • Matheus



CUES

- New block: **Block 2 / DevOps & MLOps**.
- Presenter: **Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Integration, delivery, deployment, and training automate different responsibilities.
- **CI**: integrate changes and run code, contract, and path checks.
- **Delivery**: keep a compatible change ready for an approval decision.
- **Deployment**: automatically release changes passing the defined policy.
- **CT**: train and evaluate candidates; completion is not production approval.
- *At our bank,* a connector repair can use CI/CD while retaining M1. Retraining is not required for every pipeline change.
- *Automation executes policy; it does not make acceptance criteria identical.*

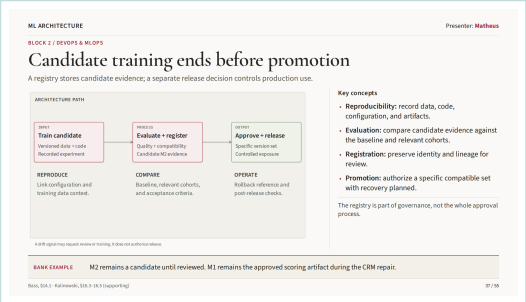
SPOKEN EXAMPLE — practice this out loud

Integration, delivery, deployment, and training automate different responsibilities. CI means integrate changes and run code, contract, and path checks. Delivery means keep a compatible change ready for an approval decision. Deployment means automatically release changes passing the defined policy. CT means train and evaluate candidates; completion is not production approval. At our bank, a connector repair can use CI/CD while retaining M1. Retraining is not required for every pipeline change. Automation executes policy; it does not make acceptance criteria identical.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

37 / 55 Candidate training ends before promotion

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A registry stores candidate evidence; a separate release decision controls production use.
- **Reproducibility**: record data, code, configuration, and artifacts.
- **Evaluation**: compare candidate evidence against the baseline and relevant cohorts.
- **Registration**: preserve identity and lineage for review.
- **Promotion**: authorize a specific compatible set with recovery planned.
- *At our bank,* m2 remains a candidate until reviewed. M1 remains the approved scoring artifact during the CRM repair.
- *The registry is part of governance, not the whole approval process.*

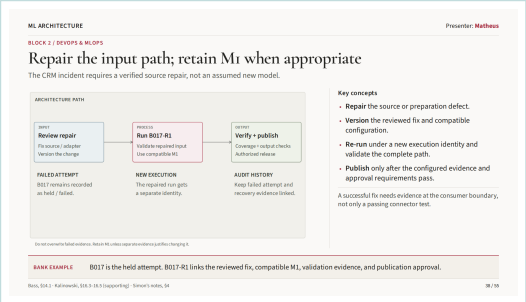
SPOKEN EXAMPLE — practice this out loud

A registry stores candidate evidence; a separate release decision controls production use. Reproducibility means record data, code, configuration, and artifacts. Evaluation means compare candidate evidence against the baseline and relevant cohorts. Registration means preserve identity and lineage for review. Promotion means authorize a specific compatible set with recovery planned. At our bank, m2 remains a candidate until reviewed. M1 remains the approved scoring artifact during the CRM repair. The registry is part of governance, not the whole approval process.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

38 / 55 Repair the input path; retain M1 when appropriate

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- The CRM incident requires a verified source repair, not an assumed new model.
- **Repair** the source or preparation defect.
- **Version** the reviewed fix and compatible configuration.
- **Re-run** under a new execution identity and validate the complete path.
- **Publish** only after the configured evidence and approval requirements pass.
- *At our bank,* b017 is the held attempt. B017-R1 links the reviewed fix, compatible M1, validation evidence, and publication approval.
- *A successful fix needs evidence at the consumer boundary, not only a passing connector test.*

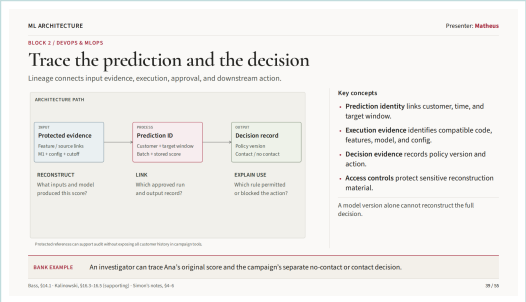
SPOKEN EXAMPLE — practice this out loud

The CRM incident requires a verified source repair, not an assumed new model. Repair means the source or preparation defect. Version means the reviewed fix and compatible configuration. Re-run means under a new execution identity and validate the complete path. Publish means only after the configured evidence and approval requirements pass. At our bank, b017 is the held attempt. B017-R1 links the reviewed fix, compatible M1, validation evidence, and publication approval. A successful fix needs evidence at the consumer boundary, not only a passing connector test.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon’s notes, §4

39 / 55 Trace the prediction and the decision

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Lineage connects input evidence, execution, approval, and downstream action.
- **Prediction identity** links customer, time, and target window.
- **Execution evidence** identifies compatible code, features, model, and config.
- **Decision evidence** records policy version and action.
- **Access controls** protect sensitive reconstruction material.
- *At our bank,* an investigator can trace Ana's original score and the campaign's separate no-contact or contact decision.
- *A model version alone cannot reconstruct the full decision.*

SPOKEN EXAMPLE — practice this out loud

Lineage connects input evidence, execution, approval, and downstream action. Prediction identity means links customer, time, and target window. Execution evidence means identifies compatible code, features, model, and config. Decision evidence means records policy version and action. Access controls means protect sensitive reconstruction material. At our bank, an investigator can trace Ana's original score and the campaign's separate no-contact or contact decision. A model version alone cannot reconstruct the full decision.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon's notes, §4–6

40 / 55 Recovery restores a compatible path

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Rehearse restoration before release, and verify it at the consumer boundary.
- **Know the target**: identify the last compatible release or validated fallback.
- **Rehearse**: test the recovery procedure before relying on it.
- **Restore the set**: include data meaning, runtime, and configuration.
- **Verify**: inspect stored scores and actual consumer behavior.
- *At our bank,* rolling M1 back cannot repair a CRM field that is still missing; the input path must be restored.
- *Handoff to security: who may read, change, approve, or operate these paths?*

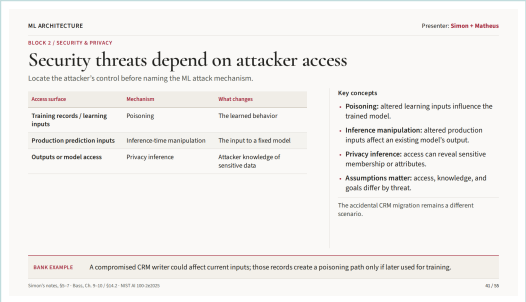
SPOKEN EXAMPLE — practice this out loud

Rehearse restoration before release, and verify it at the consumer boundary. Know the target means identify the last compatible release or validated fallback. Rehearse means test the recovery procedure before relying on it. Restore the set means include data meaning, runtime, and configuration. Verify means inspect stored scores and actual consumer behavior. At our bank, rolling M1 back cannot repair a CRM field that is still missing; the input path must be restored. Handoff to security: who may read, change, approve, or operate these paths?

Reading anchor: Breck, infrastructure tests 6–7 · Simon’s notes, §4

41 / 55 Security threats depend on attacker access

Block 2 / Security & privacy • Simon + Matheus



CUES

- New block: **Block 2 / Security & privacy**.
- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Locate the attacker's control before naming the ML attack mechanism.
- **Poisoning**: altered learning inputs influence the trained model.
- **Inference manipulation**: altered production inputs affect an existing model's output.
- **Privacy inference**: access can reveal sensitive membership or attributes.
- **Assumptions matter**: access, knowledge, and goals differ by threat.
- *At our bank,* a compromised CRM writer could affect current inputs; those records create a poisoning path only if later used for training.
- *The accidental CRM migration remains a different scenario.*

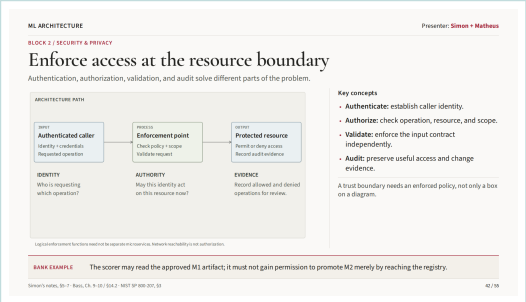
SPOKEN EXAMPLE — practice this out loud

Locate the attacker's control before naming the ML attack mechanism. Poisoning means altered learning inputs influence the trained model. Inference manipulation means altered production inputs affect an existing model's output. Privacy inference means access can reveal sensitive membership or attributes. Assumptions matter means access, knowledge, and goals differ by threat. At our bank, a compromised CRM writer could affect current inputs; those records create a poisoning path only if later used for training. The accidental CRM migration remains a different scenario.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST AI 100-2e2025

42 / 55 Enforce access at the resource boundary

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Authentication, authorization, validation, and audit solve different parts of the problem.
- **Authenticate**: establish caller identity.
- **Authorize**: check operation, resource, and scope.
- **Validate**: enforce the input contract independently.
- **Audit**: preserve useful access and change evidence.
- *At our bank,* the scorer may read the approved M1 artifact; it must not gain permission to promote M2 merely by reaching the registry.
- *A trust boundary needs an enforced policy, not only a box on a diagram.*

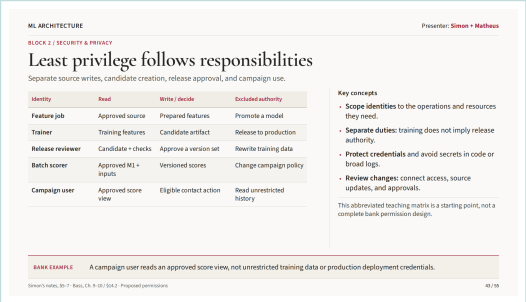
SPOKEN EXAMPLE — practice this out loud

Authentication, authorization, validation, and audit solve different parts of the problem. Authenticate means establish caller identity. Authorize means check operation, resource, and scope. Validate means enforce the input contract independently. Audit means preserve useful access and change evidence. At our bank, the scorer may read the approved M1 artifact; it must not gain permission to promote M2 merely by reaching the registry. A trust boundary needs an enforced policy, not only a box on a diagram.

Reading anchor: Simon’s notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-207, §3

43 / 55 Least privilege follows responsibilities

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Separate source writes, candidate creation, release approval, and campaign use.
- **Scope identities** to the operations and resources they need.
- **Separate duties**: training does not imply release authority.
- **Protect credentials** and avoid secrets in code or broad logs.
- **Review changes**: connect access, source updates, and approvals.
- *At our bank,* a campaign user reads an approved score view, not unrestricted training data or production deployment credentials.
- *This abbreviated teaching matrix is a starting point, not a complete bank permission design.*

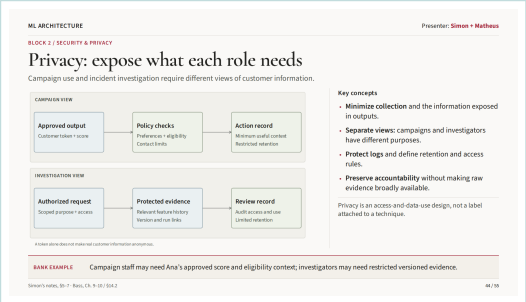
SPOKEN EXAMPLE — practice this out loud

Separate source writes, candidate creation, release approval, and campaign use. Scope identities means to the operations and resources they need. Separate duties means training does not imply release authority. Protect credentials means and avoid secrets in code or broad logs. Review changes means connect access, source updates, and approvals. At our bank, a campaign user reads an approved score view, not unrestricted training data or production deployment credentials. This abbreviated teaching matrix is a starting point, not a complete bank permission design.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · Proposed permissions

44 / 55 Privacy: expose what each role needs

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Campaign use and incident investigation require different views of customer information.
- **Minimize collection** and the information exposed in outputs.
- **Separate views**: campaigns and investigators have different purposes.
- **Protect logs** and define retention and access rules.
- **Preserve accountability** without making raw evidence broadly available.
- *At our bank,* campaign staff may need Ana’s approved score and eligibility context; investigators may need restricted versioned evidence.
- *Privacy is an access-and-data-use design, not a label attached to a technique.*

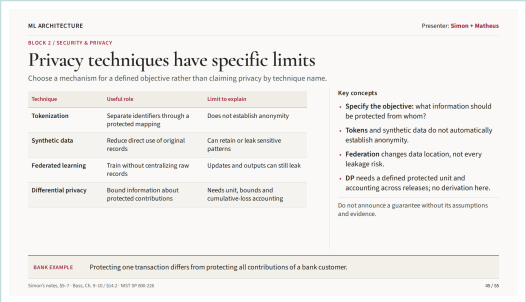
SPOKEN EXAMPLE — practice this out loud

Campaign use and incident investigation require different views of customer information. Minimize collection means and the information exposed in outputs. Separate views means campaigns and investigators have different purposes. Protect logs means and define retention and access rules. Preserve accountability means without making raw evidence broadly available. At our bank, campaign staff may need Ana’s approved score and eligibility context; investigators may need restricted versioned evidence. Privacy is an access-and-data-use design, not a label attached to a technique.

Reading anchor: Simon’s notes, §5–7 · Bass, Ch. 9–10 / §14.2

45 / 55 Privacy techniques have specific limits

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Choose a mechanism for a defined objective rather than claiming privacy by technique name.
- **Specify the objective**: what information should be protected from whom?
- **Tokens** and synthetic data do not automatically establish anonymity.
- **Federation** changes data location, not every leakage risk.
- **DP** needs a defined protected unit and accounting across releases; no derivation here.
- *At our bank,* protecting one transaction differs from protecting all contributions of a bank customer.
- *Do not announce a guarantee without its assumptions and evidence.*

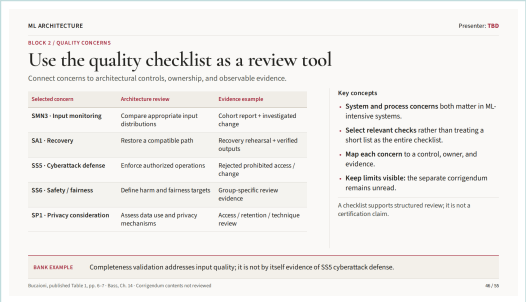
SPOKEN EXAMPLE — practice this out loud

Choose a mechanism for a defined objective rather than claiming privacy by technique name. Specify the objective means what information should be protected from whom? Tokens means and synthetic data do not automatically establish anonymity. Federation means changes data location, not every leakage risk. DP means needs a defined protected unit and accounting across releases; no derivation here. At our bank, protecting one transaction differs from protecting all contributions of a bank customer. Do not announce a guarantee without its assumptions and evidence.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-226

46 / 55 Use the quality checklist as a review tool

Block 2 / Quality concerns • TBD



CUES

- New block: **Block 2 / Quality concerns**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Connect concerns to architectural controls, ownership, and observable evidence.
- **System and process concerns** both matter in ML-intensive systems.
- **Select relevant checks** rather than treating a short list as the entire checklist.
- **Map each concern** to a control, owner, and evidence.
- **Keep limits visible**: the separate corrigendum remains unread.
- *At our bank,* completeness validation addresses input quality; it is not by itself evidence of SS5 cyberattack defense.
- *A checklist supports structured review; it is not a certification claim.*

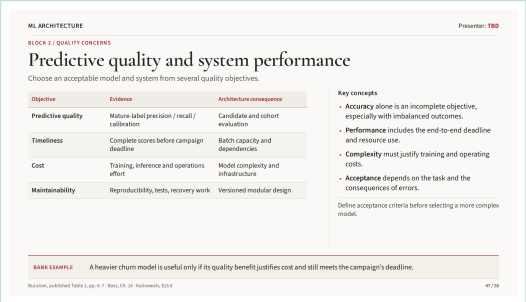
SPOKEN EXAMPLE — practice this out loud

Connect concerns to architectural controls, ownership, and observable evidence. System and process concerns means both matter in ML-intensive systems. Select relevant checks means rather than treating a short list as the entire checklist. Map each concern means to a control, owner, and evidence. Keep limits visible means the separate corrigendum remains unread. At our bank, completeness validation addresses input quality; it is not by itself evidence of SS5 cyberattack defense. A checklist supports structured review; it is not a certification claim.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Corrigendum contents not reviewed

47 / 55 Predictive quality and system performance

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Choose an acceptable model and system from several quality objectives.
- **Accuracy** alone is an incomplete objective, especially with imbalanced outcomes.
- **Performance** includes the end-to-end deadline and resource use.
- **Complexity** must justify training and operating costs.
- **Acceptance** depends on the task and the consequences of errors.
- *At our bank,* a heavier churn model is useful only if its quality benefit justifies cost and still meets the campaign's deadline.
- *Define acceptance criteria before selecting a more complex model.*

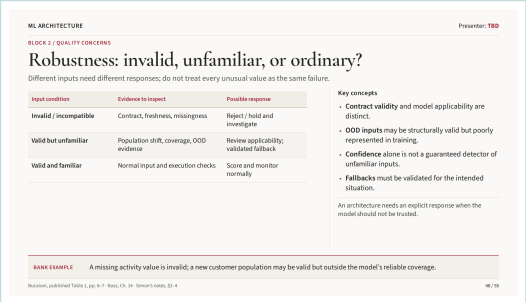
SPOKEN EXAMPLE — practice this out loud

Choose an acceptable model and system from several quality objectives. Accuracy means alone is an incomplete objective, especially with imbalanced outcomes. Performance means includes the end-to-end deadline and resource use. Complexity means must justify training and operating costs. Acceptance means depends on the task and the consequences of errors. At our bank, a heavier churn model is useful only if its quality benefit justifies cost and still meets the campaign's deadline. Define acceptance criteria before selecting a more complex model.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Kalinowski, §13.4

48 / 55 Robustness: invalid, unfamiliar, or ordinary?

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Different inputs need different responses; do not treat every unusual value as the same failure.
- **Contract validity** and model applicability are distinct.
- **OOD inputs** may be structurally valid but poorly represented in training.
- **Confidence** alone is not a guaranteed detector of unfamiliar inputs.
- **Fallbacks** must be validated for the intended situation.
- *At our bank,* a missing activity value is invalid; a new customer population may be valid but outside the model's reliable coverage.
- *An architecture needs an explicit response when the model should not be trusted.*

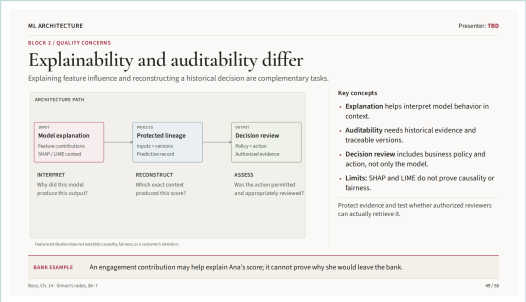
SPOKEN EXAMPLE — practice this out loud

Different inputs need different responses; do not treat every unusual value as the same failure. Contract validity means and model applicability are distinct. OOD inputs means may be structurally valid but poorly represented in training. Confidence means alone is not a guaranteed detector of unfamiliar inputs. Fallbacks means must be validated for the intended situation. At our bank, a missing activity value is invalid; a new customer population may be valid but outside the model's reliable coverage. An architecture needs an explicit response when the model should not be trusted.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Simon's notes, §2–4

49 / 55 Explainability and auditability differ

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Explaining feature influence and reconstructing a historical decision are complementary tasks.
- **Explanation** helps interpret model behavior in context.
- **Auditability** needs historical evidence and traceable versions.
- **Decision review** includes business policy and action, not only the model.
- **Limits**: SHAP and LIME do not prove causality or fairness.
- *At our bank,* an engagement contribution may help explain Ana’s score; it cannot prove why she would leave the bank.
- *Protect evidence and test whether authorized reviewers can actually retrieve it.*

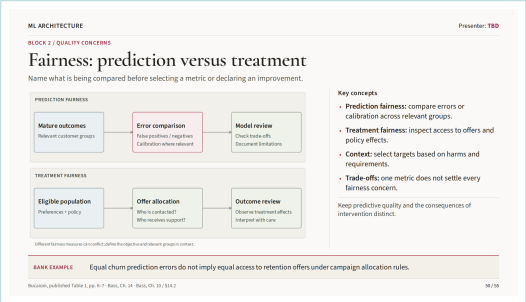
SPOKEN EXAMPLE — practice this out loud

Explaining feature influence and reconstructing a historical decision are complementary tasks. Explanation means helps interpret model behavior in context. Auditability means needs historical evidence and traceable versions. Decision review means includes business policy and action, not only the model. Limits means sHAP and LIME do not prove causality or fairness. At our bank, an engagement contribution may help explain Ana’s score; it cannot prove why she would leave the bank. Protect evidence and test whether authorized reviewers can actually retrieve it.

Reading anchor: Bass, Ch. 14 · Simon’s notes, §6–7

50 / 55 Fairness: prediction versus treatment

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Name what is being compared before selecting a metric or declaring an improvement.
- **Prediction fairness**: compare errors or calibration across relevant groups.
- **Treatment fairness**: inspect access to offers and policy effects.
- **Context**: select targets based on harms and requirements.
- **Trade-offs**: one metric does not settle every fairness concern.
- *At our bank,* equal churn prediction errors do not imply equal access to retention offers under campaign allocation rules.
- *Keep predictive quality and the consequences of intervention distinct.*

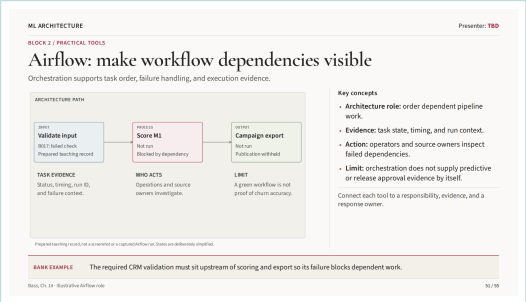
SPOKEN EXAMPLE — practice this out loud

Name what is being compared before selecting a metric or declaring an improvement. Prediction fairness means compare errors or calibration across relevant groups. Treatment fairness means inspect access to offers and policy effects. Context means select targets based on harms and requirements. Trade-offs means one metric does not settle every fairness concern. At our bank, equal churn prediction errors do not imply equal access to retention offers under campaign allocation rules. Keep predictive quality and the consequences of intervention distinct.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Bass, Ch. 10 / §14.2

51 / 55 Airflow: make workflow dependencies visible

Block 2 / Practical tools • TBD



CUES

- New block: **Block 2 / Practical tools**.
- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Orchestration supports task order, failure handling, and execution evidence.
- **Architecture role**: order dependent pipeline work.
- **Evidence**: task state, timing, and run context.
- **Action**: operators and source owners inspect failed dependencies.
- **Limit**: orchestration does not supply predictive or release approval evidence by itself.
- *At our bank,* the required CRM validation must sit upstream of scoring and export so its failure blocks dependent work.
- *Connect each tool to a responsibility, evidence, and a response owner.*

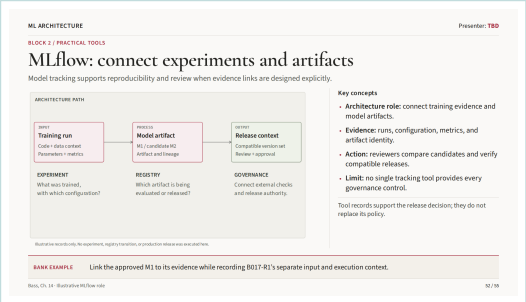
SPOKEN EXAMPLE — practice this out loud

Orchestration supports task order, failure handling, and execution evidence. Architecture role means order dependent pipeline work. Evidence means task state, timing, and run context. Action means operators and source owners inspect failed dependencies. Limit means orchestration does not supply predictive or release approval evidence by itself. At our bank, the required CRM validation must sit upstream of scoring and export so its failure blocks dependent work. Connect each tool to a responsibility, evidence, and a response owner.

Reading anchor: Bass, Ch. 14 · Illustrative Airflow role

52 / 55 MLflow: connect experiments and artifacts

Block 2 / Practical tools • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Model tracking supports reproducibility and review when evidence links are designed explicitly.
- **Architecture role**: connect training evidence and model artifacts.
- **Evidence**: runs, configuration, metrics, and artifact identity.
- **Action**: reviewers compare candidates and verify compatible releases.
- **Limit**: no single tracking tool provides every governance control.
- *At our bank,* link the approved M1 to its evidence while recording B017-R1's separate input and execution context.
- *Tool records support the release decision; they do not replace its policy.*

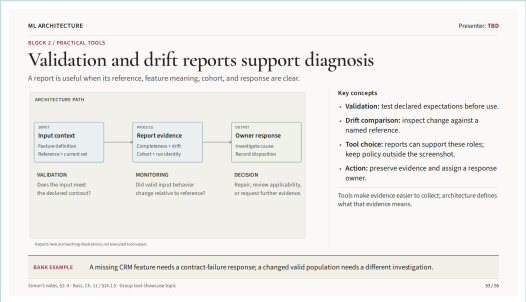
SPOKEN EXAMPLE — practice this out loud

Model tracking supports reproducibility and review when evidence links are designed explicitly. Architecture role means connect training evidence and model artifacts. Evidence means runs, configuration, metrics, and artifact identity. Action means reviewers compare candidates and verify compatible releases. Limit means no single tracking tool provides every governance control. At our bank, link the approved M1 to its evidence while recording B017-R1's separate input and execution context. Tool records support the release decision; they do not replace its policy.

Reading anchor: Bass, Ch. 14 · Illustrative MLflow role

53 / 55 Validation and drift reports support diagnosis

Block 2 / Practical tools • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- A report is useful when its reference, feature meaning, cohort, and response are clear.
- **Validation**: test declared expectations before use.
- **Drift comparison**: inspect change against a named reference.
- **Tool choice**: reports can support these roles; keep policy outside the screenshot.
- **Action**: preserve evidence and assign a response owner.
- *At our bank,* a missing CRM feature needs a contract-failure response; a changed valid population needs a different investigation.
- *Tools make evidence easier to collect; architecture defines what that evidence means.*

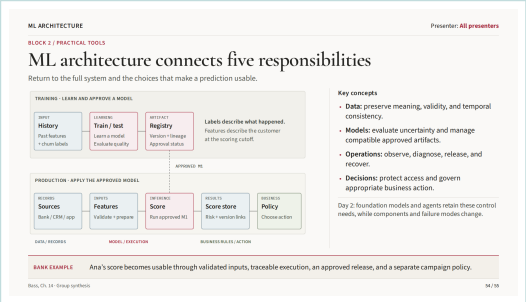
SPOKEN EXAMPLE — practice this out loud

A report is useful when its reference, feature meaning, cohort, and response are clear. Validation means test declared expectations before use. Drift comparison means inspect change against a named reference. Tool choice means reports can support these roles; keep policy outside the screenshot. Action means preserve evidence and assign a response owner. At our bank, a missing CRM feature needs a contract-failure response; a changed valid population needs a different investigation. Tools make evidence easier to collect; architecture defines what that evidence means.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Group tool-showcase topic

54 / 55 ML architecture connects five responsibilities

Block 2 / Practical tools • All presenters



CUES

- Presenter: **All presenters**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- Return to the full system and the choices that make a prediction usable.
- **Data**: preserve meaning, validity, and temporal consistency.
- **Models**: evaluate uncertainty and manage compatible approved artifacts.
- **Operations**: observe, diagnose, release, and recover.
- **Decisions**: protect access and govern appropriate business action.
- *At our bank,* ana's score becomes usable through validated inputs, traceable execution, an approved release, and a separate campaign policy.
- *Day 2: foundation models and agents retain these control needs, while components and failure modes change.*

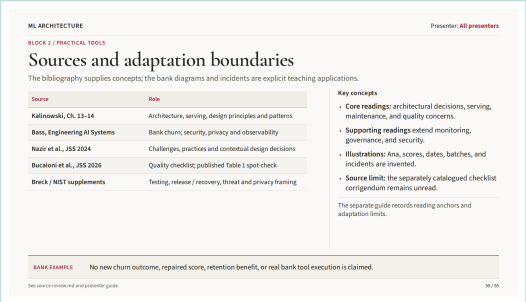
SPOKEN EXAMPLE — practice this out loud

Return to the full system and the choices that make a prediction usable. Data means preserve meaning, validity, and temporal consistency. Models means evaluate uncertainty and manage compatible approved artifacts. Operations means observe, diagnose, release, and recover. Decisions means protect access and govern appropriate business action. At our bank, ana's score becomes usable through validated inputs, traceable execution, an approved release, and a separate campaign policy. Day 2: foundation models and agents retain these control needs, while components and failure modes change.

Reading anchor: Bass, Ch. 14 · Group synthesis

55 / 55 Sources and adaptation boundaries

Block 2 / Practical tools • All presenters



CUES

- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

- The bibliography supplies concepts; the bank diagrams and incidents are explicit teaching applications.
- **Core readings**: architectural decisions, serving, maintenance, and quality concerns.
- **Supporting readings** extend monitoring, governance, and security.
- **Illustrations**: Ana, scores, dates, batches, and incidents are invented.
- **Source limit**: the separately catalogued checklist corrigendum remains unread.
- *At our bank,* no new churn outcome, repaired score, retention benefit, or real bank tool execution is claimed.
- *The separate guide records reading anchors and adaptation limits.*

SPOKEN EXAMPLE — practice this out loud

The bibliography supplies concepts; the bank diagrams and incidents are explicit teaching applications. Core readings means architectural decisions, serving, maintenance, and quality concerns. Supporting readings means extend monitoring, governance, and security. Illustrations means ana, scores, dates, batches, and incidents are invented. Source limit means the separately catalogued checklist corrigendum remains unread. At our bank, no new churn outcome, repaired score, retention benefit, or real bank tool execution is claimed. The separate guide records reading anchors and adaptation limits.

Reading anchor: See source-review.md and presenter guide